

Lecture 18: Inference for a Difference of Two Proportions

Comparing two population proportions:

Example: Do a higher proportion of male college students engage in binge drinking behavior than female students?

- We randomly sampled $n_1 = 5348$ male college students and found that $X_1 = 1392$ engage in binge drinking behavior.
- We randomly sampled $n_2 = 8471$ female college students and found that $X_2 = 1748$ engage in binge drinking behavior.

Sampling distribution for a difference of sample proportions:

Back to example: construct a 95% confidence interval for the difference in binge drinking rates for males and females.

Hypothesis testing for a difference of proportions:

Back to example: Can we conclude that male college students engage in binge drinking behavior at a different rate than female college students at a 1% level of significance?

Example: Growth in student cell phone use in the four years between 2000 and 2004:

- In 2000, 43% of a random sample of 1200 students regularly used a cell phone.
- In 2004, another sample of 1200 students was taken and 89% regularly used a cell phone.

Did cell phone use more than double in 4 years?